

Linfang Zheng

POSTDOCTORAL FELLOW · THE UNIVERSITY OF HONG KONG

1088 Xueyuan Avenue, Nanshan, Shenzhen, 518055, P.R. China

☎ (+86) 186-4655-1986 | ✉ lfzheng@hku.hk | 🏠 lynne-zheng-linfang.github.io | 🎓 Google scholar

Profile

“Postdoctoral researcher in robotics and computer vision, specializing in 6D object pose estimation, robotic manipulation, and 3D perception. Complemented by a strong background in micro-electronics and embedded hardware design, enabling cross-disciplinary expertise across both perception algorithms and integrated robotic systems.”

Education

Ph.D. in University of Birmingham (UoB)

SCHOOL OF COMPUTER SCIENCE

- Thesis title: Visual 6D Object Pose Estimation and Tracking
- Supervisors: Prof. Hyung Jin Chang, Prof. Aleš Leonardis, and Prof. Wei Zhang

Birmingham, UK

Jan. 2020 - Dec. 2024

MSc. in Harbin Institute of Technology (HIT)

INTEGRATED CIRCUIT ENGINEERING

- Dissertation title: Design and Implementation of a High-Precision Multi-Channel Lidar Measurement and Control System
- Outstanding Master's Graduate, Outstanding Master Thesis (Silver Award)

Harbin, China

Sep. 2015 - Jul. 2017

BSc. in Harbin Institute of Technology (HIT)

ELECTRONIC INFORMATION SCIENCE AND TECHNOLOGY

- Direct Admission to Post Graduate School

Harbin, China

Sep. 2011 - Jul. 2015

Employment

The University of Hong Kong (HKU)

POSTDOCTORAL FELLOW

- Leading research projects and mentoring students in robotic manipulation and vision-based perception for robotics.

Hong Kong, China

Jan. 2025 - PRESENT

Southern University of Science and Technology (SUSTech)

VISITING SCHOLAR

- Leading research projects and mentoring students in robotic manipulation and vision-based perception for robotics.

Shenzhen, China

Oct. 2024 - PRESENT

Southern University of Science and Technology (SUSTech)

RESEARCH ASSISTANT

- Assisted in project and research work including embedded software and hardware design, algorithm implementation, and project proposal writing.

Shenzhen, China

Mar. 2019 - Jan. 2020

DJI Co. Ltd.

EMBEDDED HARDWARE ENGINEER

- Worked on embedded hardware design and component evaluation, contributing to cost optimization and supply chain stability.

Shenzhen, China

Jul. 2017 - Mar. 2019

DJI Co. Ltd. (Internship)

EMBEDDED HARDWARE ENGINEER

- Supported Robomasters competition field hardware circuit-related tasks and summer camp activities.

Shenzhen, China

Jul. 2016 - Sep. 2016

Publications

- 2025 Chuye Zhang, Xiaoxiong Zhang, Wei Pan, Linfang Zheng[†], Wei Zhang[†]
Generative Visual Foresight Meets Task-Agnostic Pose Estimation in Robotic Table-Top Manipulation,
The Conference on Robot Learning (CoRL 2025) October, 2025 (Corresponding author).
- 2025 Tze Ho Elden Tse, Runyang Feng, Linfang Zheng, Jiho Park, Yixing Gao, Jihie Kim, Aleš Leonardis, Hyung Jin Chang
Collaborative Learning for 3D Hand-Object Reconstruction and Compositional Action Recognition from Egocentric RGB Videos Using Superquadrics,
The Thirty-Ninth AAAI Conference on Artificial Intelligence (AAAI 2025) February, 2025.

- 2024 Linfang Zheng, Tze Ho Elden Tse, Chen Wang, Yinghan Sun, Hua Chen, Aleš Leonardis, Wei Zhang,
GeoReF: Geometric Alignment Across Shape Variation for Category-level Object Pose Refinement,
IEEE Proc. Computer Vision and Pattern Recognition (CVPR), June, 2024.
- 2024 Yinghan Sun, Linfang Zheng, Hua Chen, Wei Zhang,
Multi-Resolution Planar Region Extraction for Uneven Terrains,
IEEE International Conference on Robotics and Automation (ICRA), May, 2024
- 2023 Linfang Zheng, Chen Wang, Yinghan Sun, Esha Dasgupta, Hua Chen, Aleš Leonardis, Wei Zhang, Hyung Jin Chang,
HS-Pose: Hybrid Scope Feature Extraction for Category-level Object Pose Estimation,
IEEE Proc. Computer Vision and Pattern Recognition (CVPR), June, 2023.
- 2023 Nora Horanyi, Linfang Zheng, Eunji Chong, Aleš Leonardis, Hyung Jin Chang
Where Are They Looking in the 3D Space?
IEEE Proc. Computer Vision and Pattern Recognition Workshop (CVPR Workshop), June, 2023. [Best Paper Award]
- 2022 Linfang Zheng, Aleš Leonardis, Tze Ho Elden Tse, Nora Horanyi, Wei Zhang, Hua Chen, Hyung Jin Chang,
TP-AE: Temporally Primed 6D Object Pose Tracking with Auto-Encoders,
IEEE International Conference on Robotics and Automation (ICRA), May, 2022
- 2022 Zhongqun Zhang, Wei Chen, Linfang Zheng, Aleš Leonardis, Hyung Jin Chang,
Trans6D: Transformer-Based 6D Object Pose Estimation and Refinement,
ECCV Workshop, 7th International Workshop on Recovering 6D Object Pose, October, 2022
- 2020 Hua Chen, Linfang Zheng, Wei Zhang
Optimal Control Inspired Q-Learning for Switched Linear Systems
American Control Conference (ACC), July, 2020

Research Experience

“My research primarily focuses on robotic manipulation and 6D object pose estimation. I also collaborate on related topics such as gaze estimation and depth completion, applying machine learning and 3D computer vision techniques to enhance machine-environment interaction.”

Optimization-based Multi-Step Cloth Pushing Planning

SUSTech & HKU, China

ROBOTICS · PLANNING · MANIPULATION · DEFORMABLE OBJECT

Dec. 2023 - PRESENT

- Robotic manipulation for clothes. **On-going.**

Depth Completion for Transparent and Specular Surfaces

HKU, China

DEPTH COMPLETION · TRANSPARENT OBJECT PERCEPTION · REAL-WORLD ROBOTICS

Dec. 2024 - PRESENT

- Co-led the development of a depth completion framework for robotic manipulation, addressing transparent and specular surfaces using affine-invariant priors and vision foundation models (**On-going**).

Structured Spatial Understanding for Vision-Language-Guided Robotic Manipulation

HKU, China

ROBOTIC MANIPULATION · SPATIAL UNDERSTANDING · VISION-BASED CONTROL · CROSS-EMBODIMENT · FEW DEMONSTRATION

Dec. 2024 - Oct. 2025

- Leading the development of a modular framework for vision-language-conditioned robotic manipulation, focusing on spatial reasoning and cross-embodiment generalization.
- Submitted to the IEEE Robotics and Automation Letters (**RAL**).

Video Generation Guided Robotic Manipulation

HKU&SUSTECH, China

ROBOTICS · VISION-BASED CONTROL · MANIPULATION · CROSS-EMBODIMENT

Dec. 2024 - May 2025

- Supervised the development of a closed-loop vision-based manipulation framework combining generative video prediction and task-agnostic pose estimation for scalable robot control.
- Accepted to the 2025 Conference on Robot Learning (**CoRL 2025**) (corresponding author).

Hand-Object Pose Estimation

UoB, UK

COMPUTER VISION · 6D OBJECT POSE · HAND POSE · GENERALIZABILITY · TRANSFORMER

Dec. 2023 - Aug. 2024

- Co-authored a study introducing superquadric-based representations to improve generalization in hand-object action recognition and object reconstruction.
- Accepted to the Thirty-Ninth AAAI Conference on Artificial Intelligence (AAAI-25) (**AAAI 2025**) (co-author).

Category-level 6D Object Pose Refinement

UoB, UK & SUSTech, China

COMPUTER VISION · 6D OBJECT POSE · CATEGORY-LEVEL · REFINEMENT · GRAPH CONVOLUTION

Mar. 2023 - Mar. 2024

- Tackled the underexplored challenge of geometric discrepancies among category-level objects in 6D pose refinement.
- Proposed a 3D graph convolution-based framework with learnable affine transformations, improving generalization and outperforming baselines using only 4% of training data (+10.5% on 10°2cm metric).
- Accepted to IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR 2024**) (first author).

Multi-Resolution Planar Region Extraction for Uneven Terrains

SUSTech, China

ROBOTICS · COMPUTER VISION · SEGMENTATION · PLANE EXTRACTION · LOCOMOTION

Nov. 2022 - Sep. 2023

- Collaboratively Introduced a multi-resolution planer region extraction strategy for uneven terrains from point cloud data.
- Contributed to designing deep learning-based plane segmentation.
- Accepted to IEEE International Conference on Robotics and Automation (ICRA 2024) (co-author).

Category-level Object Pose Estimation

UoB, UK & SUSTech, China

COMPUTER VISION · 6D OBJECT POSE · CATEGORY-LEVEL · GRAPH CONVOLUTION · BACKBONE · COMPLEX-SHAPED OBJECT

Mar. 2022 - May 2023

- Proposed the HS-layer, a novel 3D graph convolution module for hybrid-scope feature extraction from point clouds, encoding scale and translation while being robust to outliers.
- Integrated HS-layer into a category-level pose estimation framework, achieving strong improvements on complex-shaped objects (+14.5% on 5°2cm metric).
- Accepted to IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2023) (first author).

3D Joint Gaze Estimation

UoB, UK

COMPUTER VISION · 3D JOINT GAZE ESTIMATION

Jan. 2022 - Apr. 2023

- This research addressed the previously unexplored problem of integrating a depth prior and a 3D joint field-of-view probability map to estimate attention targets in a scene.
- I collaboratively introduced the cutting-edge depth-aware joint attention estimation framework, surpassing current benchmarks.
- Accepted to IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshop (CVPR Workshop 2023) (co-author).

Instance-level 6D Object Pose Tracking

UoB, UK & SUSTech, China

COMPUTER VISION · 6D OBJECT POSE · TRACKING · SEVERE OCCLUSION · TEXTURELESS OBJECT · GRU · AUTO-ENCODER

Feb. 2020 - Mar. 2022

- Proposed the first neural network-based prior pose generation method for instance-level 6D object pose tracking, leveraging pose history to improve robustness in challenging scenarios like textureless and occluded objects.
- Developed a real-time, temporally-primed architecture for pose estimation, significantly improving accuracy and robustness on benchmark datasets without sacrificing real-time performance.
- Accepted to IEEE International Conference on Robotics and Automation (ICRA 2022) (first author).

Instance-level Object Pose Estimation and Refinement

UoB, UK

COMPUTER VISION · 6D OBJECT POSE · ESTIMATION · REFINEMENT · TRANSFORMER

Aug. 2021 - Mar. 2022

- Collaboratively introduced a Transformer-based network, leveraging global feature correlation to enhance object pose estimation performance.
- Accepted to European Conference on Computer Vision Workshop (ECCV Workshop 2022) (co-author).

Optimal Control Inspired Q-Learning for Switched Linear Systems

SUSTech, China

OPTIMAL CONTROL · REINFORCEMENT LEARNING · Q-LEARNING · SWITCHED LINEAR SYSTEM

May 2019 - Dec. 2019

- Collaboratively proposed an algorithm with a carefully designed parametric approximator that respects the analytical structure of the exact Q-function, paired with an associate parameter training algorithm.
- Accepted to American Control Conference (ACC 2020) (co-author).

Honors & Awards

2023	Best Paper Award , CVPR Workshop on GAZE 2023	Vancouver, Canada
2017	Outstanding Master's Graduate , Harbin Institute of Technology	Harbin, China
2017	Silver Award for Outstanding Master's Thesis , Harbin Institute of Technology	Harbin, China
2017	First Prize Scholarship , Harbin Institute of Technology (2011 - 2017)	Harbin, China
2014	Second Prize , HIT Technology Innovation and Entrepreneurship Training Program	Harbin, China
2014	Second Prize , HIT First Physics Academic Competition	Harbin, China
2014	Second Prize Scholarship , People's Daily (People.cn)	Harbin, China

Skills

Programming	Python, C, Matlab, Verilog
Deep Learning	PyTorch, TensorFlow
Hardware Design	Embedded Hardware Design, Integrated Circuit Design, FPGA
Software Design	Embedded Software Design
Languages	English, Mandarin (Mother Language)

Peer-Review Activity

Conference

NuerIPS, CVPR, ECCV, ICCV, ICRA, IROS, CASE, AAAI, CoRL